MIL-P-45172F(AR)
AMENDMENT 2
18 October 1977

SUPERSEDING
AMENDMENT 1
6 December 1974

MILITARY SPECIFICATION

PROJECTILE, 20MM, API, M53, METAL PARTS

This amendment forms a part of Military Specification MIL-P-45172F(MU), 22 December 1972, and is approved for use by all Departments and Agencies of the Department of Defense.

Para 2.1, under SPECIFICATIONS add the following:

"MILITARY

MIL-I-6868 - Inspection Process, Magnetic Particle"

Para 3.4 Soundness, delete the entire paragraph and substitute the following:

"3.4 Soundness,

3.4.1 Soundness Body. -The body, Dwg. D7258807, shall be free of ferrous metal discontinuity detectable by magnetic particle inspection.

3.4.2 Soundness Adapter. -Metal discontinuity in the adapter of the body adapter and rotating band assembly, as determined by eddy current inspection, shall be less than that represented by Production Check Standard (PCS) No. 1."

Para 4,3.3.2, Body, in third line delete Critical: None defined, and substitute the following:

"Critical:
1. Ferrous metal discontinuity ---------- Soundness test (see 4.4.11)"

* Para 4.3.3.6 under Minor, delete the following:

"204. Base radius missing ---------- Visual"

* Para 4.3.3.7 under Major, add the following:

"104. Radius, base of projectile ---------- Visual/Gage 5/

* Under note 4/ add the following:

FSC 1305

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"5/ Measurement of the radius at base of projectiles shall apply where visual inspection indicates that the radius dimension is borderline."

Para 4.3.4, Test, add the following test at the end of Table I and footnote at end of notes to Table 1:

"Test	Sample Size	Item	Rqmt. Para.
Soundness Verification 12/	315	Body	3.4.1

12/ If ferrous metal discontinuity in excess of 0.005 inch in width (excluding knurl surface(s) is found in any unit of the sample as witnessed by the test of 4.4.11, the lot shall be rejected."

Para 4.4.4, delete title "Soundness" and substitute "Soundness (Adapter Only)."

After para 4.4.10, add the following:

"4.4.11 Soundness Body. -The method of inspection shall be in accordance with MIL-I-6868 for fluorescent suspension, utilizing the wet, residual, circular method and a magnetizing current of 400 + 50 amperes. (Liquid. -Water with a suitable inhibitor and wetting agent may be used as a liquid vehicle without approval from the procuring activity.) Electrodes shall make contact with the base end and the tip of each body. Except for the knurled area, the entire ferrous metal surface of each body of the lot shall be examined for magnetic particle indications of discontinuity."

"4.4.11.1 Soundness verification. -The inspection sample shall be processed as prescribed in 4.4.11. Each body of the sample showing magnetic particle indication(s) of ferrous metal discontinuity judged to be in excess of government established visual standards shall be sectioned 90 degrees to the longitudinal axis at the approximate midpoint of maximum magnetic particle concentration. A sample body which shows ferrous metal discontinuity that measures more than 0.005 inch in width shall be classed defective."

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Preparing Activity:

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